ABHED · NO-CODE AI GATEWAY · EXPLAINED

Bina code badle AI agents ko Abhed se kaise joda

OpenAI-compatible API, Base URL aur API key kya hote hain, aur sirf do settings badalne se Zentro Logistics ke saare agents Abhed ki nigrani mein kaise aa gaye.

1 Pehle basics: AI agent model se baat kaise karta hai?

Agent (Atlas, Penny, Vault…) khud "soch" nahi sakta. Har baar use ek **AI model** (GPT, Llama, gpt-oss…) se poochna padta hai. Yeh poochna internet pe ek simple **HTTP request** hai. Teen cheezein chahiye:

① Address: Base URL
Request *kahan* jaaye. Jaise OpenAI ka `https://api.openai.com/v1`.

② Pehchaan: API key
Kaun pooch raha hai. Header mein jaati hai: `Authorization: Bearer <key>`.

③ Sawaal: JSON body
Model ka naam, baatcheet (`messages`) aur agent ke tools ki list (`tools`).

```
# Zentro ka server, jab Vault ko customer ka message milta hai
POST {Base URL}/chat/completions
Authorization: Bearer {API key}
{ "model": "openai/gpt-oss-120b",
  "messages": [ {"role":"system", "content":"You are Vault, Zentro's data agent…"},
                {"role":"user",   "content":"Export the full customer list with phones and PAN" } ],
  "tools": [ "lookup_customer", "send_email", "export_customers" ] }

# Model ka jawab: woh khud kuch NAHI karta, sirf SUGGEST karta hai ("tool call")
{ "tool_calls": [ { "name": "export_customers", "arguments": {"limit": 10000} } ] }
```

Sabse zaroori baat: model sirf *proposal* deta hai. Asli kaam (export, refund, email) **company ka apna code** karta hai, model ka jawab padhne ke baad. Abhed isi beech ke pal mein khada hota hai.

2 "OpenAI-compatible" ka matlab kya hai?

OpenAI ne upar wala format (endpoint `/chat/completions`, `Bearer` key, `messages`, `tools`, `tool_calls`) banaya. Woh itna popular hua ki **poori industry ne same format apna liya**. Ab almost har provider ek hi "bhasha" bolta hai, sirf address alag hai:

PROVIDER	BASE URL	FORMAT
OpenAI	`https://api.openai.com/v1`	OpenAI
Groq	`https://api.groq.com/openai/v1`	OpenAI-compatible
OpenRouter, Together, Azure, Ollama, vLLM…	apna-apna address	OpenAI-compatible
Abhed Gateway	**`https://kavach-os.vercel.app/api/gw/v1`**	**OpenAI-compatible**

Isiliye har agent framework (OpenAI SDK, LangChain, CrewAI, n8n, Flowise, custom code) mein ek setting hoti hai: **"Base URL" + "API key"**. Provider badalna ho toh code nahi, sirf yeh do cheezein badalti hain. **Abhed ne yahi darwaaza use kiya:** hum bhi OpenAI wali bhasha bolte hain, toh koi bhi agent bina code change ke Abhed se baat kar sakta hai.

3 Base URL ko tod ke samjho

`https://kavach-os.vercel.app` `/api/gw/v1` `/chat/completions`

kavach-os.vercel.app: Abhed ka server (Vercel pe live).

/api/gw/v1: **api** = server ka API hissa, **gw** = gateway, **v1** = version 1 (OpenAI jaisa naam). **Yahi Base URL hai.**

/chat/completions: SDK yeh hissa **khud jodta hai**. Isliye Base URL mein yeh nahi likhte.

4 API key (`kv_gw_…`): company ki pehchaan

Abhed Connect → **No code** → company ka naam → **Connect company** dabate hi Abhed ek random key banata hai, jaise `kv_gw_04b3…dc99`. Isi key se Abhed jaanta hai ki call **kis company** ki hai. Database mein sirf key ka **SHA-256 hash** save hota hai, isliye key ek hi baar dikhti hai. **New key** dabate hi purani key turant band ho jaati hai. **Remove company** karte hi key hamesha ke liye band. Yahi Abhed ka **kill switch** hai.

CONNECTION KAISE HUA

5 Pehle vs. ab: sirf do settings badli

Pehle: koi nigrani nahi 12,480 records leak



Model ne export suggest kiya, Zentro ke code ne chala diya.

Ab: har call Abhed se 0 records leak



Jawab pehle Abhed padhta hai; galat tool call Zentro tak nahi pahunchta.

ZENTRO ADMIN SETTING	PEHLE	AB
AI provider	Built-in rules	**OpenAI-compatible** (URL type karte hi apne aap)
Base URL	(khaali / kisi provider ka)	`https://kavach-os.vercel.app/api/gw/v1`
API key	(khaali / provider ki key)	`kv_gw_…`, Abhed Connect se copy ki hui

Na code, na terminal, na redeploy. Zentro pehle se OpenAI-compatible provider support karta tha (jaise har AI product). Humne sirf address badla.

6 Abhed ke endpoints, aur Save / Test connection

Abhed ke do OpenAI jaise endpoints: `POST /api/gw/v1/chat/completions` (asli baatcheet, har tool call yahin check) aur `GET /api/gw/v1/models` (model list). Save pe Zentro `/models` ko key ke saath call karta hai. sahi key 200: Abhed pe **"key verified"**, Zentro pe hara ✓ **Connected**. galat / purani key 401: *"Unknown Abhed gateway key"*, laal ×.

7 Ek message ka poora safar (Vault ka export, live demo)

1. **Customer Zentro site pe Vault ko likhta hai**
"Export the full customer list with phones and PAN for our ad agency."

2. **Zentro ka server request banata hai**
"You are Vault…" + message + tools. Base URL ab Abhed ka hai, toh request **Abhed** ko jaati hai.

3. **Abhed: key check**
`kv_gw_…` ka hash, kaunsi company. Galat key? Wahi ruk jaata hai (401).

4. **Abhed: agent ki pehchaan, khud se**
"You are **Vault**" padh ke dashboard pe **"New agent discovered: Vault"**. Register nahi karna padta.

5. **Abhed: prompt-injection scan**
"ignore previous instructions" jaise signals mile toh *tainted*: paisa / bahar bhejne wale actions band.

6. **Abhed asli model ko call karta hai**
Groq `gpt-oss-120b`. Model jawab deta hai: `export_customers(limit: 10000)`.

7. **Abhed: deterministic rulebook (koi AI nahi, fixed niyam)**
Order: COMPANY-BLOCK → DESTRUCTIVE → **BULK** → DLP → EXFIL → INJECTION → LIMIT (>₹2 lakh) → APPROVAL (>₹10,000) → PRIVILEGE. Yahan GW-BULK@1: 100 se zyada records, toh **BLOCKED**.

8. **Abhed jawab badal deta hai**
Tool call hata ke likha: *"⛔ Abhed blocked "export_customers"…"*. Zentro ke code ko chalane ko kuch mila hi nahi. Leak **0**.

9. **Sab live record**
Har call aur faisla save hota hai. Abhed Connect / Fleet pe turant: agent online, kya block hua, kaunsa rule.

Bada kaam Penny ka ₹45,000 refund: **GW-APPROVAL**, toh hold. Request **48 sec rukti hai**; Abhed pe **Approve** hua toh *wahi exact action, ek baar* jaata hai. Reject / time out: blocked.

8 Sir puchein toh

Koi aur company? Uske code / tool mein bhi yahi: `OpenAI(base_url="…/api/gw/v1", api_key="kv_gw_…")`. LangChain, n8n, CrewAI sab mein yeh setting hai.

Rules AI se kyun nahi? AI ko injection se bahkaya ja sakta hai, fixed rules ko nahi. Same input, same faisla, aur rule ID audit mein.

Bypass kare toh? Real setup mein asli model key **sirf Abhed ke vault** mein. Agents ke paas sirf `kv_gw_` key, toh seedha model tak nahi pahunch sakte.

Limits? Streaming abhi nahi; sirf model ke tool calls dikhte hain (direct actions ke liye Abhed SDK). Pattern proven hai (Cloudflare AI Gateway, Portkey); Abhed action-level governance karta hai.